

Chunking and Chunk Size

Chunking divides a long document into smaller text passages that can be processed and retrieved independently. A chunk may contain several sentences, a paragraph, or a short section. Its size is commonly measured in tokens, although some applications use characters or words.

Long documents often cover many topics and may exceed an embedding model's input limit. Representing an entire document as one unit can hide a small but relevant detail inside unrelated material. Splitting the document gives a retriever more focused passages to choose from.

Chunk size controls how much information each passage contains. A practical strategy respects paragraph boundaries and keeps headings with their explanations. For example, a maintenance manual can be split into coherent procedures, allowing a question about replacing a filter to retrieve the relevant instructions.

Overlap and Retrieval Quality

Chunk overlap repeats a portion of text between neighboring chunks. This can preserve continuity when an explanation or instruction crosses a split boundary. For example, repeating the final sentence of one passage at the beginning of the next may keep a condition connected to the action it qualifies.

Chunks that are too small can lose definitions, context, and relationships between ideas. Chunks that are too large can mix topics and make relevant details harder to distinguish. Excessive overlap also increases storage and can produce nearly duplicate search results.

Chunking affects retrieval quality because it determines the units that a search system can select. Test several sizes and overlap settings with representative questions. Prefer boundaries that preserve complete ideas, and inspect whether retrieved chunks contain enough context to answer accurately.